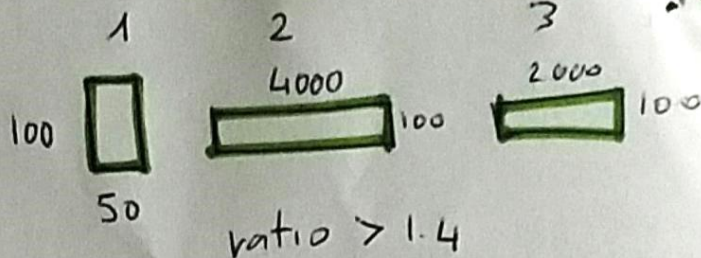
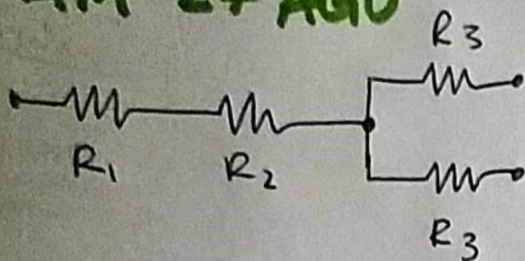


EXAM 27 AGO

1)



$$R = \frac{\eta \cdot L}{12} \cdot \left(1 - 0.63 \frac{\text{short}}{\text{long}}\right) \cdot \text{short}^3 \cdot \text{long}$$

$$R_1 = \frac{12 \cdot 10^{-3} \cdot L_1}{(1 - 0.63 \cdot \frac{1}{2}) \cdot (50 \mu\text{m})^3 \cdot 100 \mu\text{m}}$$

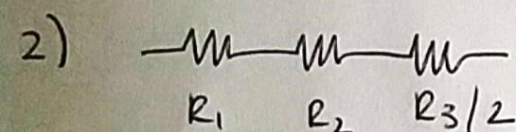
$$7 \cdot 10^{12} \frac{\text{Pa} \cdot \text{s}}{\text{m}^3}$$

$$R_2 = \frac{12 \cdot 10^{-3} \cdot L_2}{(100 \mu\text{m})^3 \cdot 4000 \mu\text{m}}$$

$$1.5 \cdot 10^{11} \frac{\text{Pa} \cdot \text{s}}{\text{m}^3}$$

$$R_3 = \frac{12 \cdot 10^{-3} \cdot L_3}{(100 \mu\text{m})^3 \cdot 2000 \mu\text{m}}$$

$$6 \cdot 10^{10} \frac{\text{Pa} \cdot \text{s}}{\text{m}^3}$$

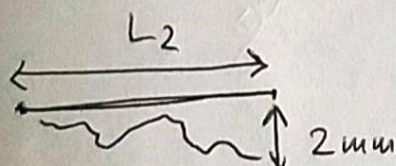


$$V_1 = \frac{Q_{IN}}{H \cdot W_1} = \frac{10^{-6} \frac{\text{L}}{\text{min}} \cdot \frac{10^{-3}}{60}}{100 \mu\text{m} \cdot 50 \mu\text{m}} = 3.3 \frac{\mu\text{m}}{\text{s}}$$

$$V_2 = \frac{Q_{IN}}{H \cdot W_2} = 41.6 \frac{\mu\text{m}}{\text{s}}$$

$$V_3 = \frac{Q_{IN}/2}{H \cdot W_3} = V_2$$

3)



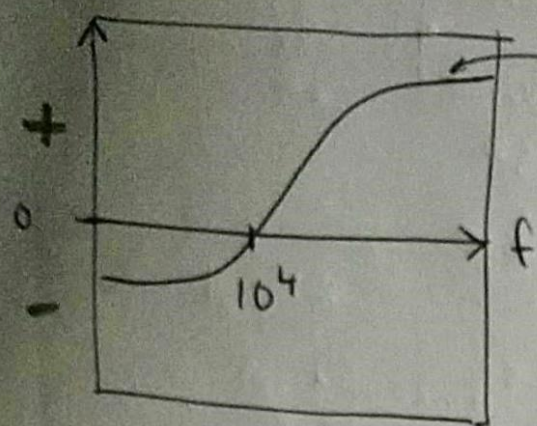
$$v = 50 \frac{\text{mm}}{\text{s}} \quad \Delta t = \frac{L_2}{v} = 1 \text{ s}$$

$$L = \sqrt{D \cdot \Delta t} = 2 \text{ mm}$$

$$D = \frac{L^2}{\Delta t} = \frac{4 \cdot 10^{-6}}{1} = 4 \cdot 10^{-6} \frac{\text{m}^2}{\text{s}}$$

$$4 \cdot 10^{-6} \frac{\text{m}^2}{\text{s}}$$

NEGATIVE DEP, $F_{DEP} \propto V^3$

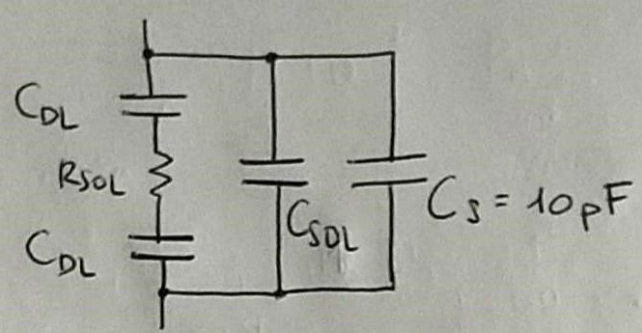
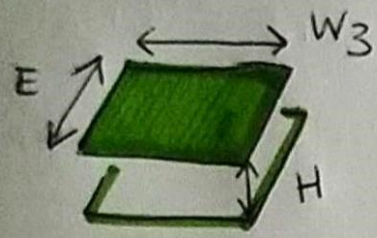


$IN1 \Rightarrow \sigma = 2 S/m$

$|CM(f)| < 0$ to be negative

$f < \sim 10^4 Hz$ (take some margin)

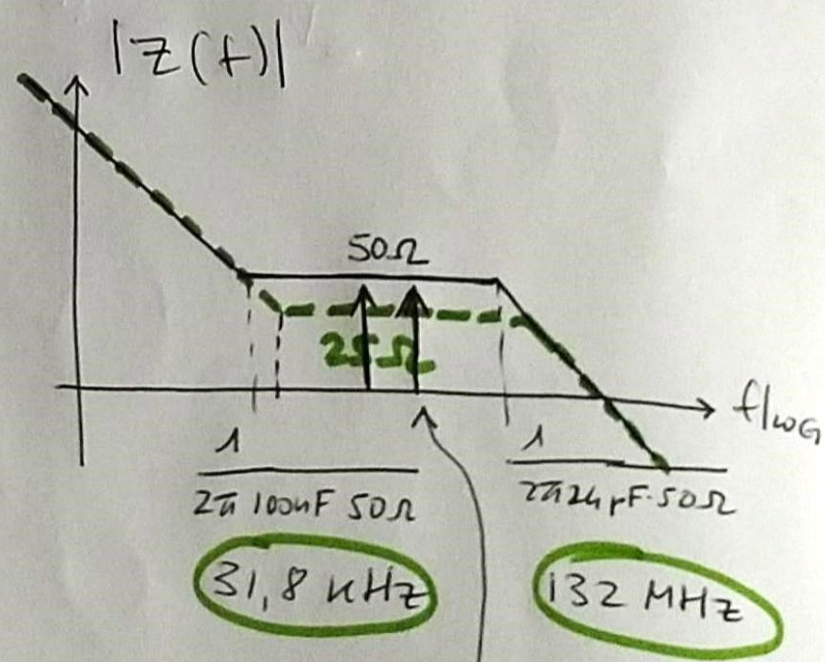
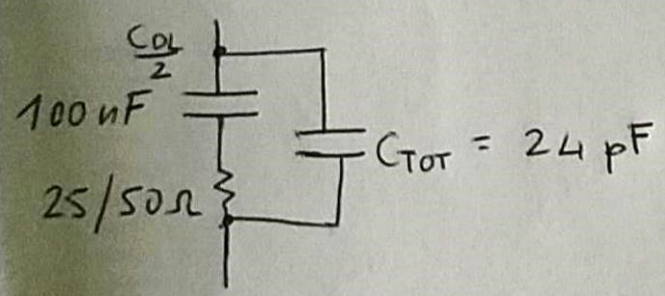
LIFT-OFF



$C = 0.1 \frac{pF}{\mu m^2} \cdot E \cdot W_3 = 0.1 \frac{pF}{\mu m^2} \cdot 1000 \cdot 2000 = 200 nF$

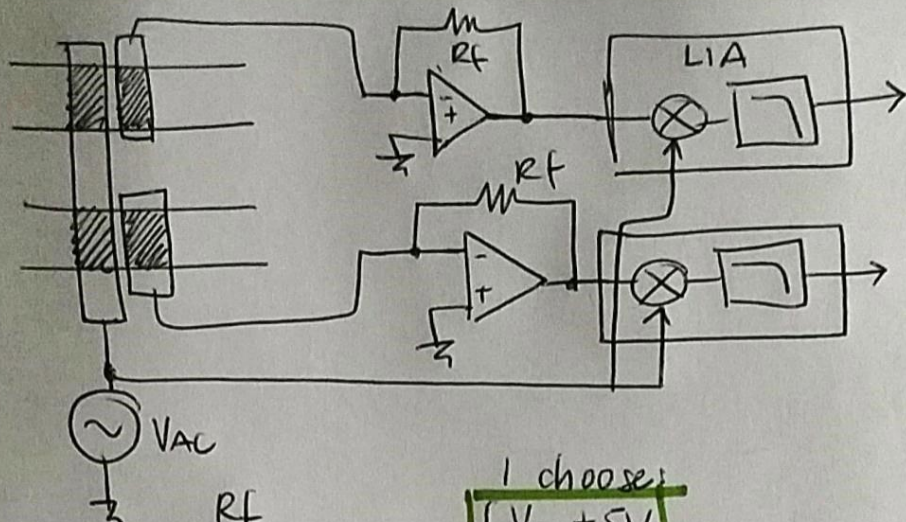
$R = \rho \cdot \frac{H}{E \cdot W_3} = \frac{1}{\sigma} \cdot \frac{H}{E \cdot W_3}$
 $50 \Omega (\sigma = 1 S/m)$
 $25 \Omega (\sigma = 2 S/m)$

$C = \epsilon_0 \epsilon_r \cdot \frac{E \cdot W_3}{H} = 14 pF$
 $\epsilon_r = 80$

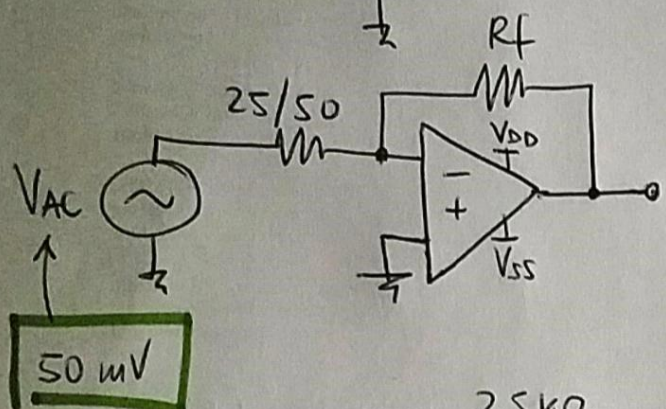


OPTIMAL FREQUENCY
 IN THE PLATEAU
 $\sim 600 kHz$

8)

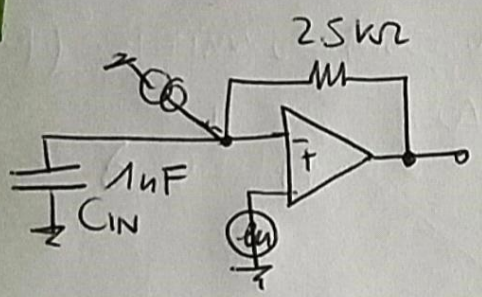


I choose:
 $V_{DD} + 5V$
 $V_{SS} - 5V$



$$R_f = \frac{5V}{50mV} \cdot 25\Omega = 2.5k\Omega$$

10)



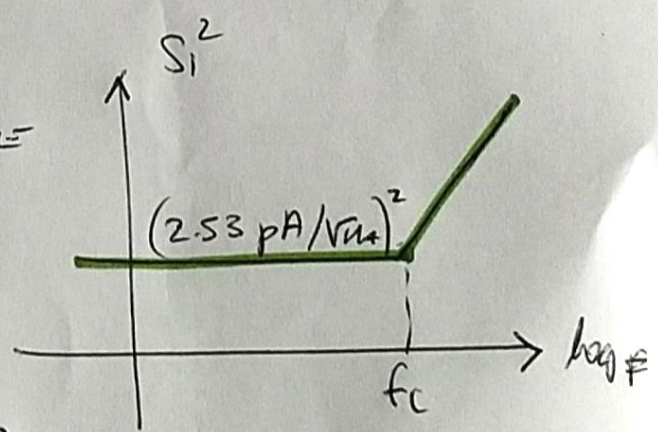
$$S_{i_{TOT}}^2 = \sqrt{\frac{4kT}{R_f} + i_n^2 + \frac{V_n^2}{R_f^2} + V_n^2(2\pi f C_{IN})^2}$$

$$\frac{4kT}{R_f} = \frac{4 \cdot 4T \cdot 9}{R_f} = (2.5 pA/\sqrt{Hz})^2$$

$$i_n^2 = (10 fA/\sqrt{Hz})^2 \text{ NEGUGIBLE}$$

$$\frac{(1\mu V)^2}{(2.5k\Omega)^2} = (0.4 pA/\sqrt{Hz})^2$$

$$\left(\frac{1\mu V}{\sqrt{Hz}} \cdot 2\pi \cdot f \cdot 1\mu F\right)^2 = (2.53 pA/\sqrt{Hz})^2$$



$$f_c = 402 kHz$$

11) $BW \sim \frac{1}{\Delta t} \sim 1 kHz$

12) DIFFERENTIAL SENSING OR MEASURE TEMP.